

Lineare Algebra
Serie 27
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1

Trying to find eigenvectors

$$v_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

$$Av_1 = \begin{pmatrix} 27 \\ -27 \\ 0 \end{pmatrix} = 27v_1$$

$$v_2 = \begin{pmatrix} 1 \\ 1 \\ ? \end{pmatrix}$$

$$Av_2 = \begin{pmatrix} 1+8? \\ 1+8? \\ 16-7? \end{pmatrix}$$

$$\Rightarrow 1+8x = \frac{16-7x}{x}$$

$$8x^2 + 8x - 16 = 0 \Rightarrow x_{1,2} \in \{1, -2\}$$

$$A \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 9 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$A \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} = -15 \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$$

The three vectors we found form an orthogonal basis for $\mathbb{R}^3$

$$u_i = \frac{v_i}{||v_i||} = \begin{cases} \frac{\sqrt{2}}{2} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \\ \frac{\sqrt{3}}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \\ \frac{\sqrt{6}}{6} \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} \end{cases}$$

$\Rightarrow$

$$P = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & -\frac{\sqrt{6}}{3} \end{pmatrix}$$
$$P^{-1} = P^T$$

2

2.a

Since $F : \mathbb{R}^4 \rightarrow \mathbb{R}^4$

injective $\Leftrightarrow$ surjective $\Leftrightarrow$ bijective

$$\forall v, w \in \mathbb{R}^4 : s(F(v), F(w)) = s(v, w)$$

Let $F(v) = 0$

$$\begin{aligned} s(F(v), F(v)) &= s(0, 0) = s(v, v) \\ &\Rightarrow v = 0 \end{aligned}$$

This implies that

$$\ker(F) = \{0\} \Leftrightarrow F \text{ is injective} \Rightarrow F \text{ bijective}$$

2.b

2.b.i

$$\text{Let } M = \begin{pmatrix} T & 0 \\ 0 & \pm 1 \end{pmatrix} \quad T \in O(3) \quad v = \begin{pmatrix} u \\ t \end{pmatrix} \in \mathbb{R}^4 \quad u \in \mathbb{R}^3$$

$$Mv = Tu \pm t$$

$$\text{Set } w = \begin{pmatrix} u' \\ t' \end{pmatrix} \quad u' = \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}$$

$$\begin{aligned} s(F(v), F(w)) &= s(Mv, Mw) \\ &= (Tu)^T(Tu') - (\pm t)(\pm t') \\ &= u^T T^T T u' - tt' \\ &= uu' - tt' \\ &= xx' + yy' + zz' - tt' \\ &= s(v, w) \end{aligned}$$

2.b.ii

$$B := \begin{pmatrix} \gamma & & -v\gamma \\ & 1 & \\ -v\gamma & & \gamma \end{pmatrix}$$

$$\forall v, w \in \mathbb{R}^4$$

$$s(v, w) := v^T \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \\ & & & -c \end{pmatrix} w$$

We want to show that it doesn't change under Lorentz transformation.

$$\begin{aligned} s(Bv, Bw) &= (Bv)^T \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \\ & & & -c \end{pmatrix} (Bw) \\ &= v^T B \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \\ & & & -c \end{pmatrix} Bw \end{aligned}$$

$$\begin{aligned}
B \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & -c \end{pmatrix} B &= \begin{pmatrix} \gamma & & & vc\gamma \\ & 1 & & \\ & & 1 & \\ -v\gamma & & & -c\gamma \end{pmatrix} \\
B \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & -c \end{pmatrix} B &= \begin{pmatrix} \gamma & & & vc\gamma \\ & 1 & & \\ & & 1 & \\ -v\gamma & & & -c\gamma \end{pmatrix} \begin{pmatrix} \gamma & & & -v\gamma \\ & 1 & & \\ & & 1 & \\ -v\gamma & & & \gamma \end{pmatrix} \\
&= \begin{pmatrix} \gamma^2(1-v^2c) & & & \gamma^2v(c-1) \\ & 1 & & \\ & & 1 & \\ \gamma^2v(c-1) & & & \gamma^2(v^2-c) \end{pmatrix} =
\end{aligned}$$

Since $c = 1$

$$\Rightarrow \dots = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & -1 \end{pmatrix}$$

Therefore

$$s(Bv, Bw) = v^T \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & -1 \end{pmatrix} w = s(v, w)$$

2.c

let φ be a linear isometry with $\det(\varphi) = 1$

Define the set C

$$v \in \mathbb{R}^4$$

$$C = \{v \in M \mid s(v, v) = 0\} \Leftrightarrow x^2 + y^2 + z^2 = ct^2$$

Define

$$A = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & -1 \end{pmatrix}$$

Since φ is an isometry. Let P be the representative Matrix of φ .

$$P^T A P = A$$

$$P^T = A P^{-1} A$$

$$P^{-1} = A P^T A$$

Since $P \in M_{4 \times 4}(\mathbb{R})$ It follows that if $\lambda \in \mathbb{C}$ is an eigenvalue than $\frac{1}{\lambda}$ is one to.

Let v be an eigenvector of λ

$$s(Pv, Pv) = \lambda^2 s(v, v) = s(v, v)$$

$$\Rightarrow (\lambda^2 - 1)s(v, v) = 0$$

if $\lambda \neq \pm 1$

$$\Rightarrow s(v, v) = 0$$

v is an eigenvector in the light cone.

if $\lambda = \pm 1$

?

3

4

4.a

Since A is orthogonal every $|\lambda| = 1$.

Since $p_A(t)$ is third order at least 1 eigenvalue has to be in $\mathbb{R}$.

$$\Rightarrow \lambda_1 = \pm 1$$

but

$$\overline{p_A(\lambda)} = p_A(\overline{\lambda}) = \overline{0} = 0$$

$\Rightarrow$ the complex eigenvalues appear pairwise

$$\det(A) = \lambda_1 \lambda_2 \lambda_3 = \lambda \overline{\lambda} \lambda_1 = |\lambda|^2 \lambda_1 = \lambda_1 = 1$$

4.b

let $v \in \mathbb{R}^3$

$$Av = v$$

$$(A - I)v = 0$$

$$(a_{11} - 1)v_1 + a_{12}v_2 + a_{13}v_3 = 0$$

$$a_{21}v_1 + (a_{22} - 1)v_2 + a_{23}v_3 = 0$$

$$a_{31}v_1 + a_{32}v_2 + (a_{33} - 1)v_3 = 0$$

We can use the fact

$$\|r_1 \times r_2\| = \sin(\theta) \|r_1\| \cdot \|r_2\|$$

$$\Rightarrow v = \begin{pmatrix} a_{11} - 1 \\ a_{12} \\ a_{13} \end{pmatrix} \times \begin{pmatrix} a_{21} \\ a_{22} - 1 \\ a_{23} \end{pmatrix} = \begin{pmatrix} a_{12}a_{23} - a_{13}(a_{22} - 1) \\ a_{13}a_{21} - (a_{11} - 1)a_{23} \\ (a_{11} - 1)(a_{22} - 1) - a_{21}a_{12} \end{pmatrix}$$

4.c

The matrix A is a rotation over $\text{span}(v)$ from exercise b. Therefore, there exists a matrix P invertible such that R has the form

$$A = P^{-1}RP$$

$$\text{with } R = \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Since $\text{tr}(ABC) = \text{tr}(BCA)$

$$\text{tr}(A) = \text{tr}(P^{-1}RP) = \text{tr}(PP^{-1}R) = \text{tr}(R) = 2\cos(\theta) + 1$$

$$\Rightarrow \cos(\theta) = \frac{\text{tr}(A) - 1}{2}$$